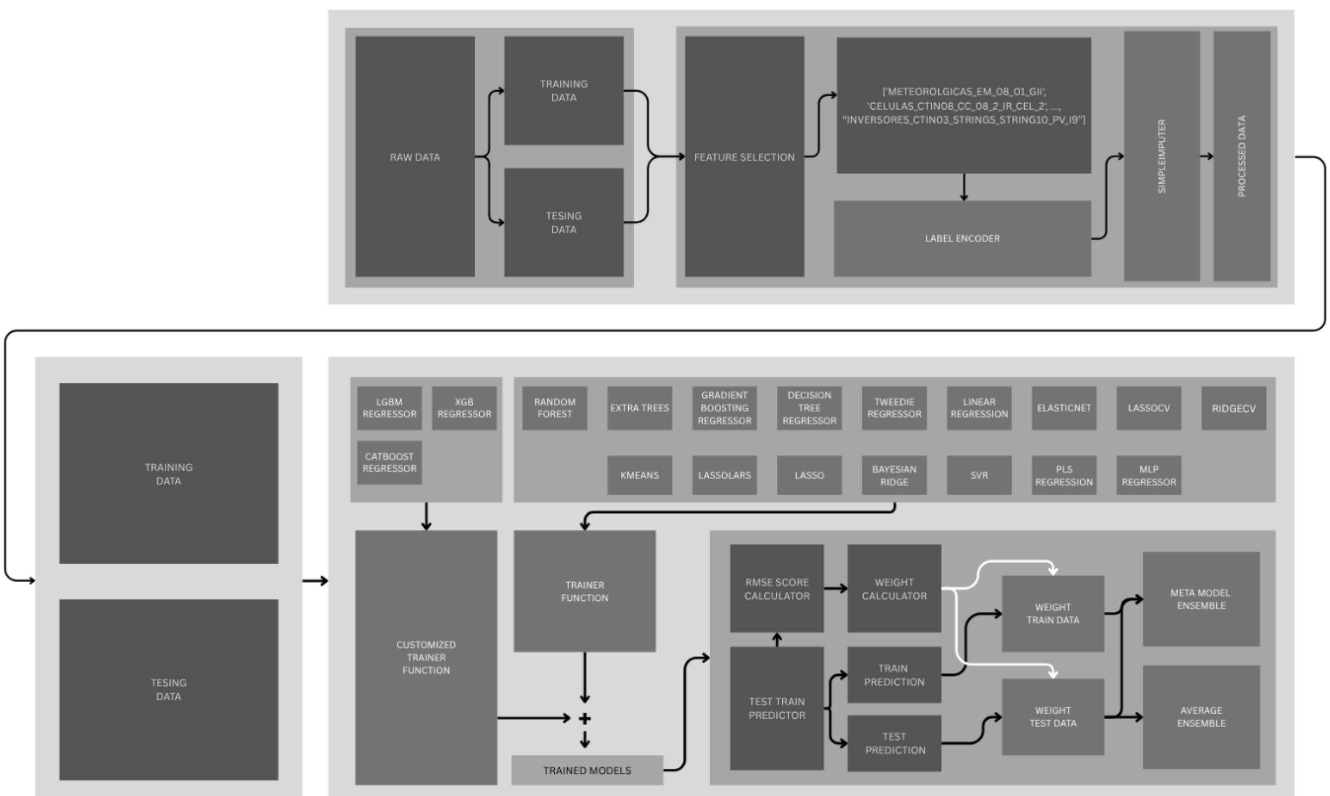
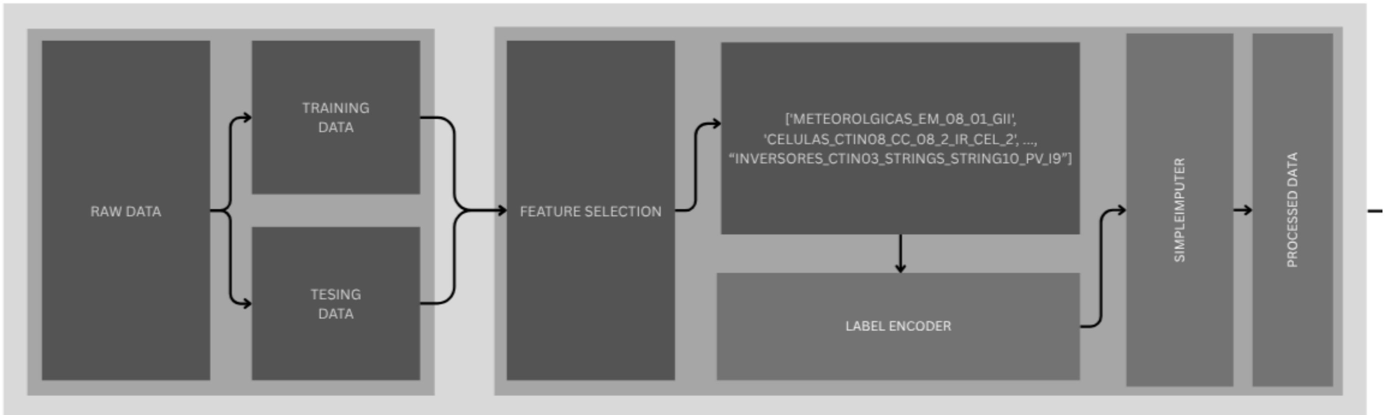
Zelestra Hackathon: Phase 2

Theoretical Modelling for Power Produced – Plant Level

APPROACH-1

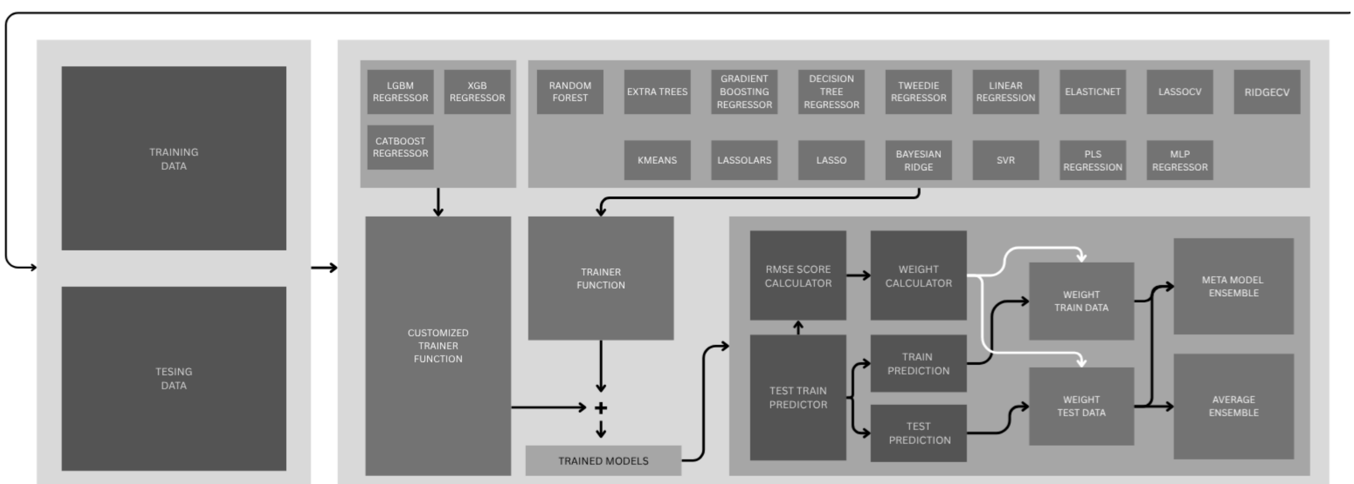
ML: REGRESSION, ENSEMBLE, STACK REGRESSION, VARIOUS AUTOML, WEIGHTED BLEND





1. **Data Preprocessing**

- Raw data is split into training and testing datasets.
- Feature selection is applied to retain relevant meteorological and inverter/cell features.
- Label encoding is used to handle categorical variables.
- Smart imputation addresses missing values.
- The processed data is prepared for model trainin**g.**



2. **Model Training**

- A wide variety of regression models are trained, including:
  - Tree-based models: LightGBM, XGBoost, CatBoost, Random Forest, Extra Trees, Gradient Boosting, Decision Tree

- Linear models: Linear Regression, ElasticNet, Lasso, LassoLars, Ridge, Bayesian Ridge, PLS Regression
- Other regressors: Tweedie, SVR, MLP Regressor
- Clustering-based features via KMeans

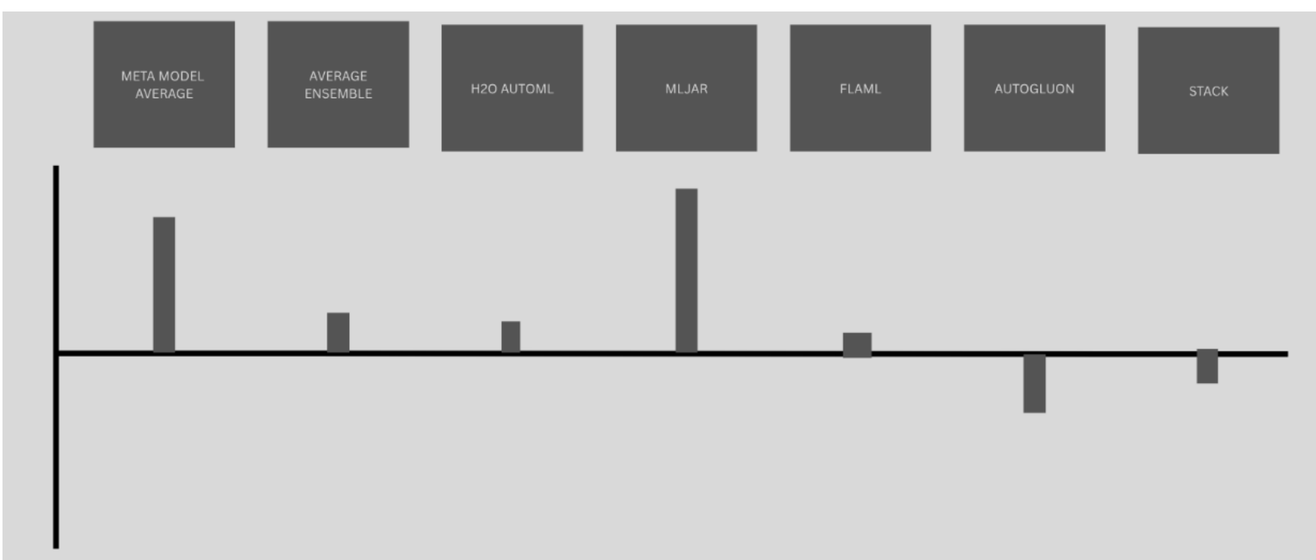
- A customized trainer function manages the training of these models.

3. Ensemble Strategy

- Individual model predictions are combined using:
    - Stacked Meta-Model Ensemble: Trains a meta-learner on the base model outputs, using weighted predictions from training data.
    - Average Ensemble: Averages predictions from multiple base models for final output.

4. Evaluation

- The pipeline includes:
    - Score calculators (e.g., RMSE)
    - Weight calculators for dynamic weighting of model outputs
    - Predictions for both training and testing sets
    - Final weighted ensemble predictions



Weighted Blend

Purpose: Combine predictions from multiple ensemble or AutoML frameworks using weighted averages to improve overall model performance and robustness.

Components:
The bars show the contribution (weight) assigned to each method in the final blended prediction:

- Meta Model Average – assigned significant weight, indicating strong performance.
- Average Ensemble – contributes moderately.
- H2O AutoML – small positive weight.
- MLJAR – large weight, another key contributor.
- FLAML – small positive weight.
- AutoGluon – negative weight (to correct bias or overfitting in its predictions).
- Stack – small negative weight, for similar adjustment.

Max Achieved Metrics:

R^2 Score: 0.9969

RMSE Score(out of 100): 30.788811799144444

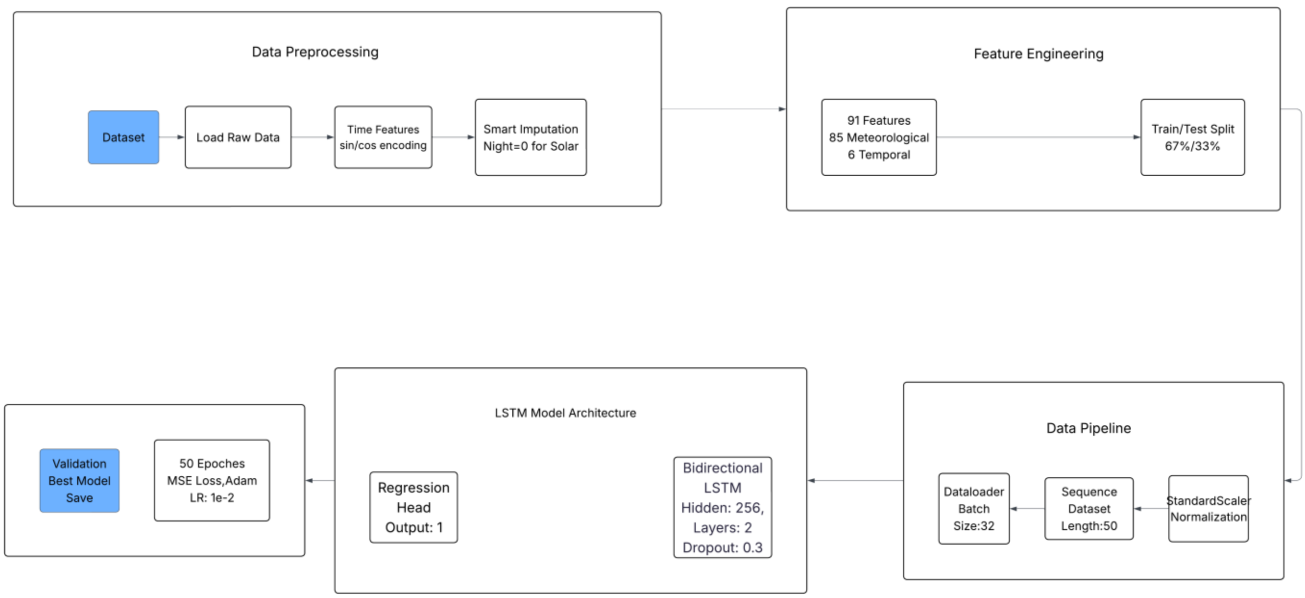
```
config={
    "auto_gluon":0,
    "M_AVERAGE":0,
    "META_MODEL":0,
    "STACK":0,
    "Tabnet":0,
    "H2O":0,
    "MLJAR":1,
    "FLAML":0
}

predict(config)
```

```
R² Score: 0.9967839943121966
```

Approach-2

Deep Learning: LSTM Time Series



1. Data Preprocessing

- The raw dataset is loaded.
- Time features are encoded using sine and cosine transformations.
- Smart imputation is applied, especially for handling solar data during night-time periods.

2. Feature Engineering

- A total of 91 features are generated: 85 meteorological and 6 temporal features.
- The dataset is split into 67% training and 33% testing sets.

3. Data Pipeline

- StandardScaler normalization is applied to the dataset.
- The data is structured into sequences of length 50.
- A dataloader is configured with a batch size of 32.

4. LSTM Model Architecture

- A bidirectional LSTM is used with:

   - Hidden size: 256
   - Number of layers: 2
   - Dropout: 0.3
- A regression head outputs a single value.

5. Training and Validation

- The model is trained for 50 epochs.
- Uses MSE loss and the Adam optimizer with a learning rate of 1e-2.
- The best model is saved based on validation performance.

Max Achieved Metrics:

R² Score: 0.9937

```
In [14]: r2_score(y_true[49:],results['predicted_power'])

Out[14]: 0.9936546920341792
```